

Performance Ranking (Eval Rate)

Rank Model		Eval Rate	Tokens Generated	Total Duration	Load Time	Quantization
1	tinyllama:1.1b	43.66 t/s	1,177	28.1 s	0.86 s	Q4_0
2	llama3.2:3b	14.38 t/s	1,176	1 m 26 s	3.38 s	Q4_K_M
3	phi3:3.8b	12.37 t/s	1,264	1 m 46 s	3.52 s	Q4_0
4	qwen2.5-coder:7b	7.19 t/s	782	1 m 49 s	0.13 s	Q4_K_M
5	qwen2.5:7b	7.13 t/s	724	1 m 57 s	14.3 s	Q4_K_M

Quality Ranking (Essay Quality)

Rank	Model	Scientific Accuracy	Structure & Coherence	Hallucinations	Overall Writing Quality	Notes
1	qwen2.5:7b	Excellent	Excellent	Low	Highest	Cleanest, most accurate Rayleigh explanation
2	qwen2.5-coder:7b	Very Good	Very Good	Low	Very High	Slightly more technical tone
3	llama3.2:3b	Good	Good	Moderate	Good	Solid, readable, minor historical inaccuracies
4	phi3:3.8b	Good	Good but wordy	Moderate	Good	Correct science, but more verbose and

Rank	Model	Scientific Accuracy	Structure & Coherence	Hallucinations	Overall Writing Quality	Notes
						slightly less polished
5	tinylama:1.1b	Poor	Poor	High	Poor	Many factual errors

Key Observations

1. Speed vs Quality Trade-off is clear

- tinylama is extremely fast but unusable for serious work.
- The two 7B Qwen models are the smartest but ~6× slower than tinylama and ~2× slower than the 3B models.
- **llama3.2:3b** sits in a very attractive middle ground (14.4 t/s + good quality).

2. llama3.2:3b vs phi3:3.8b

- Almost identical speed (14.4 vs 12.4 t/s).
- llama3.2 feels slightly more natural and better structured.
- phi3 is a bit more verbose and academic-sounding.

3. qwen2.5:7b vs qwen2.5-coder:7b

- Virtually the same speed (~7.1–7.2 t/s).
- Regular qwen2.5:7b produced the cleanest and most accurate essay.
- The coder version is still excellent, just slightly more technical in tone.

4. Your earlier visual impression was correct

- llama3.2:3b and the two 7B Qwen models are **not dramatically different** in writing quality on this type of task.
- The biggest jump in quality is from tinylama → everything else.
- The jump from 3B/3.8B → 7B is real but smaller than the speed penalty.

Recommendation for Your “Agent Sweet Spot”

Given your hardware (Ryzen 7 5700X + RX 580 8GB in Docker):

Priority	Model	Why
Best overall	llama3.2:3b	Best balance of speed (14.4 t/s) + quality. Strong candidate for daily agent use.
Strong alternative	phi3:3.8b	Very close to llama3.2, slightly slower, good reasoning.
When quality matters most	qwen2.5:7b	Highest quality, but only ~7 t/s. Use when accuracy is more important than speed.
Avoid for agents	tinyllama:1.1b	Too weak.